

6.6. Statistics Basics



Why Statistics Matters in ML?



About Statistics

Statistics helps us understand, summarise, and interpret data before modeling.

Purpose in ML

- Identify patterns in data
- Detect issues like missing values or noise
- Improve model performance by preparing clean features

Example

Are most values centered tightly or widely spread?

Central Tendency - Overview



Central Tendency

Shows the central or typical value in data. Represents the “center of gravity” of your dataset.

Purpose in ML

Helps understand where most values lie. Useful for summarising large datasets.

Example

Student test scores, daily temperatures.

Mean, Median & Mode



Mean

Average of all values.

$$(10 + 20 + 30) / 3 = 20$$

Use in ML: Used for imputing missing values.



Median

Middle value when sorted.

Values: 5, 8, 11 -> 8

Use in ML: Robust when data has outliers.



Mode

Most frequent value.

Values: 2, 2, 3 -> 2

Use in ML: Useful for categorical features.

Measure of Spread - Overview



Measure of Spread

Measure of Spread tells how far values vary from the center.

Importance in ML

High spread indicates unstable (noisy) or inconsistent features.
Low spread implies stable and predictable data

Example

Values between 20 and 40 have higher spread than values between 28 and 32.

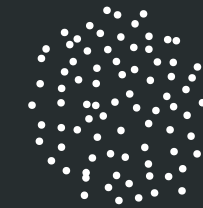
Range, Variance, SD & IQR

Range



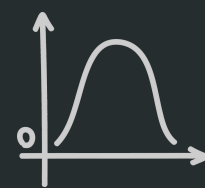
Max minus min. Fast way
to see spread.

Variance



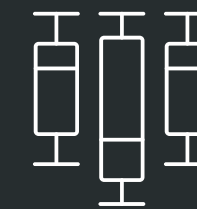
Shows how far values
deviate from the mean
on average.

Standard Deviation



Square root of variance.
Low SD means values are stable.
Use in ML: Feature scaling

Inter Quartile Range



Middle 50 percent of the data
(Q1 to Q3).
Use in ML: Outlier detection

Distribution

Normal Distribution

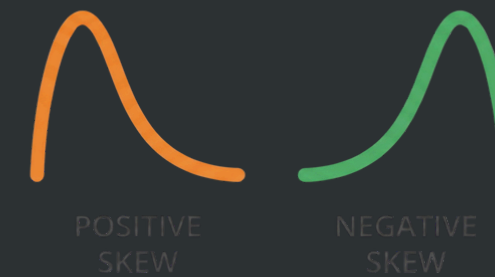


Symmetrical bell curve

Example: Human heights

Use in ML: Scaling and assumptions in linear models.

Skewed Distribution



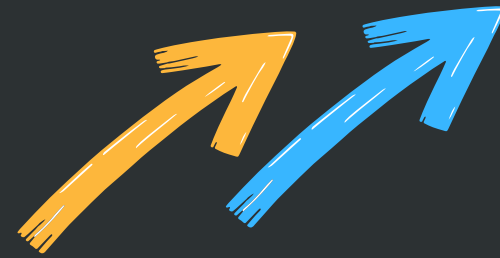
More data on one side

Example: Salaries

Use in ML: Suggests using log transformations.

Covariance and Correlation

Covariance



Shows how two variables move together.

Example: Height and weight often increase together.

Correlation



Strength and direction of relationship.

Range: -1 to 1

Use in ML: Feature selection, removing multicollinearity

Percentiles and Quartiles



Percentile

Shows value below which a percentage of data lies.
Example: 90th percentile means better than 90 percent of values.

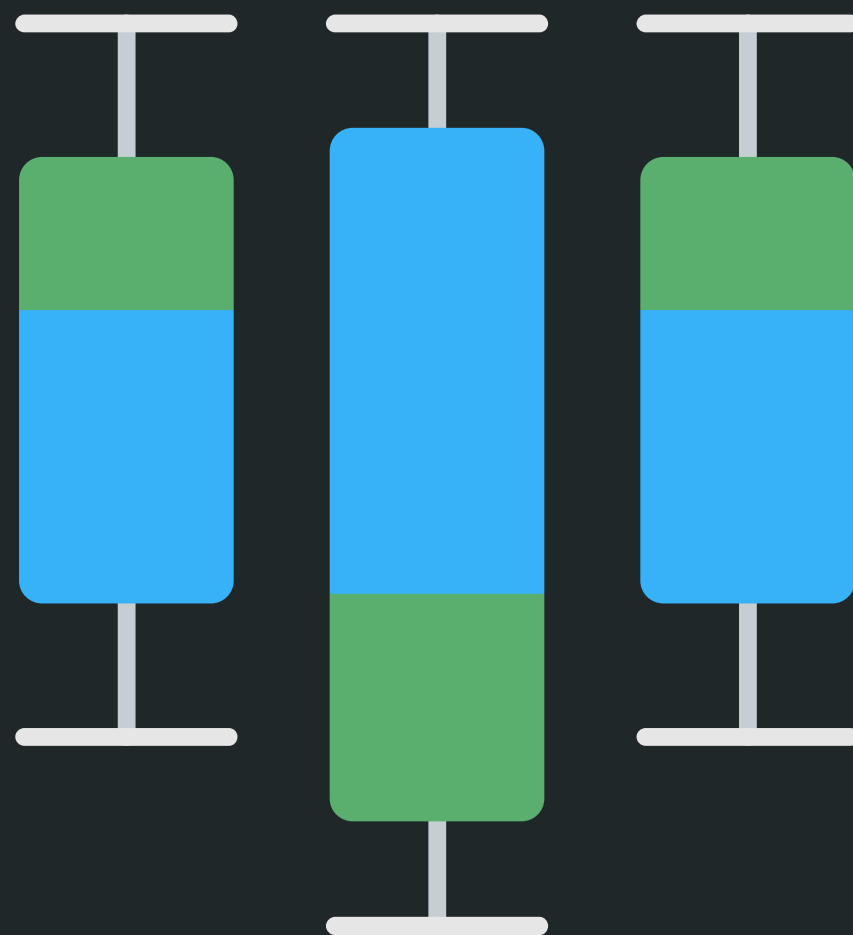
Quartile

- Q1: 25 percent
- Q2: 50 percent (median)
- Q3: 75 percent

Use in ML

Understand distribution and detect outliers.

Outliers



Outlier

Values far from most other values.

Example

Salary 200000 in a dataset of 20000 to 40000.

Detection

IQR method (1.5 times IQR rule)
 $IQR = Q3 - Q1$

Why Important in ML

- Skews regression line
- Affects mean and variance
- Can slow or confuse gradient descent

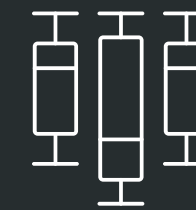
How Statistics Is Used in ML

EDA



Find patterns,
distributions, issues

Outlier Treatment



Clean noisy data.

Scaling



Standardisation & normalisation

Evaluation



Understand error spread.